



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP2523 – DATABASE

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INITIAL SYSTEM PROPOSAL

**SYSTEM *KADA's Cooperative
System***

Group Name: *Group 5*

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Stakeholder: *Lembaga Kemajuan Pertanian Kemubu
(KADA)*

Representative Name	Date Interviewed
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PROPOSED SYSTEM BACKGROUND

1. Introduction

This proposal introduces a digital solution aimed at transforming KADA's cooperative membership process, from initial application to final approval. The system will create an efficient digital platform where applicants, administrators, and Board Members (ALK) can seamlessly interact, ensuring faster processing times and greater transparency throughout the membership and loan processes.

Lembaga Kemajuan Pertanian Kemubu (KADA) is a cooperative organization that supports agricultural development through financial assistance and services. However, its current manual system for managing membership registrations and loan applications is slow, prone to errors, and inconvenient for members and staff. This results in processing delays, risks of data loss, and challenges in managing an increasing number of members.

To address these challenges, the KADA Cooperation System is proposed as a digital platform to streamline processes. The system will allow applicants to register for membership online. Once approved, members can apply for loans, track their application status, and calculate repayment plans using a built-in loan calculator. Admins will manage data and verify applications, while Board Members (ALK) will review and decide on loan approvals.

This system aims to enhance operational efficiency, improve data security, and provide members with easier access to services. Developed using the Waterfall Methodology, the project will progress through structured phases to meet KADA's specific needs. This initiative will modernize KADA's operations, reduce manual workloads, and position the organization for future growth.

2. The Existing System

a. Background of existing system

The cooperation system used by KADA is currently being done manually. All of the forms and paperwork are submitted and stored in physical by the client and the organization. Applicants need to attend the KADA office to obtain the application form, and there is no computerized system for the submission of the documents required. The application type is divided into two, member application and loan application. If a client wants to apply for a loan, one must first become a member of KADA cooperation. The applicant needs to submit a member application and pay RM50 to become a member. When the application for membership is approved, then only they can proceed with loan application. Applicant's information is kept in a physical paper and the clerk must check their application one by one. Then, board members of KADA will conduct a meeting weekly to verify the application and decide whether to accept or reject the application. Once the result is finalized, the applicant will be notified through SMS messages. KADA offers a various type of loan for its member including Al-Bai, Al-Innah, Vehicle Restoration, Road Tax & Insurance and Al-Qadrul Hassan. Applicants can choose what type of loan they want to apply for according to their needs. The processes for loan application are quite similar to member application where applicant must come to the KADA office and obtain the application form, thus submit it with all the documents related at the office physically. A meeting between board members will be conducted also to determine the decision for the application. Lastly, successful applicants will be notified through SMS messages.

In addition, KADA's cooperation system includes a shares system which all members are eligible to join. Members can contribute to the organization's capital through purchasing of shares. These shares represent their ownership in the organization. Upon first time to apply for membership, the applicant needs to pay RM50. The registration fee will then be recorded as the applicant's shares when they are successful in becoming a member. Additionally, the shares amount will also put into factors which the member is eligible to apply for loan or not, all members must have at least RM350 worth of shares for them to apply for loan. If their shares do not fulfill the condition, they must first purchase the shares until they reach the required minimum amount.

b. Issues with existing system

The issues arising from the manual handling of the application which are:

- i) **Data Security & Reliability** – All the related documents containing sensitive information about customers were stored physically. There might be risk of losing or leaking the data to malicious party because of human error. Moreover, the organization of the documents might not be in a systematic order, and it is possible

for the data to mistakenly match with the wrong customer and end up providing misinformation about the customer. Thus, understanding the importance of having a convenient system in an organization might help the organization to provide secure and more reliable data for the customer.

- ii) **Timeliness** – The process of requiring and submitting documents is time consuming as the applicant needs to travel to KADA office multiple times. It requires a large amount of time just for the enquiries and submission of the documents. Some of the members might be living far away across the country and the distance between KADA office and their hometown must be put into consideration. To put it simply, the organization must provide a centralized system for the documentation which is accessible by the customer whenever or wherever they want.
- iii) **Scalability** – As time grows, KADA might experience an increasing number of its members. The manual system that is currently being used may struggle to handle a large volume of requests and applications in the future. Besides, the weekly meetings for applications approval might become a bottleneck and lead to slower response to the applicant, which will frustrate applicants. In summary, KADA needs to computerize their systems to give the best customer experience and enhance the efficiency of the system.

3. The Proposed System

The new system for KADA Cooperation is a digital portal designed to simplify and streamline membership and loan applications processes. Applicants can apply for membership online, and once approved, they become members and gain access to loan services. Admins manage and verify applications, while Board Members (ALK) are responsible for making final loan approval decisions. The system improves efficiency, provides secure data storage, and enhances the overall user experience.

Overview of the proposed system

The system will serve four types of users:

1. Applicants:

- Individuals applying for KADA Cooperation membership
- Applicants must submit an online application form and pay the required membership fee
- They receive real-time updates about their application status

2. Members:

- Once approved, applicants become members and can apply for loans anytime
- Members can use the loan calculator to estimate repayment plans and track their loan application statuses.

3. Admins:

- Admins handle all administrative tasks, including verifying membership and loan applications.
- They update applications statuses and forward approved loan applications to the Board Members (ALK) for final decisions.
- Admins also manage user data and ensure smooth operation of the system.

4. Board Members (ALK):

- Board Members review loan applications forwarded by Admins.
- They make the final decision on loan approvals or rejection based on eligibility criteria and KADA policies.

Key Features:

- Online membership registration for applicants.
- Loan application and tracking for members.
- Real-time updates on application status via email notifications.
- A loan calculator to help members plan repayment schedules.
- A centralized system for admins and board members to manage and review applications.

Mission Objectives

1. To make KADA's membership and loan applications processes faster and more efficient.
2. To provide a secure and user-friendly platform accessible to applicants, members, admins and board members.
3. To automate routine tasks like status updates and data management, reducing manual errors.
4. To streamline decision-making processes for loan approvals.
5. To support KADA's operational goals with a scalable and efficient system.

System Scope and Boundaries

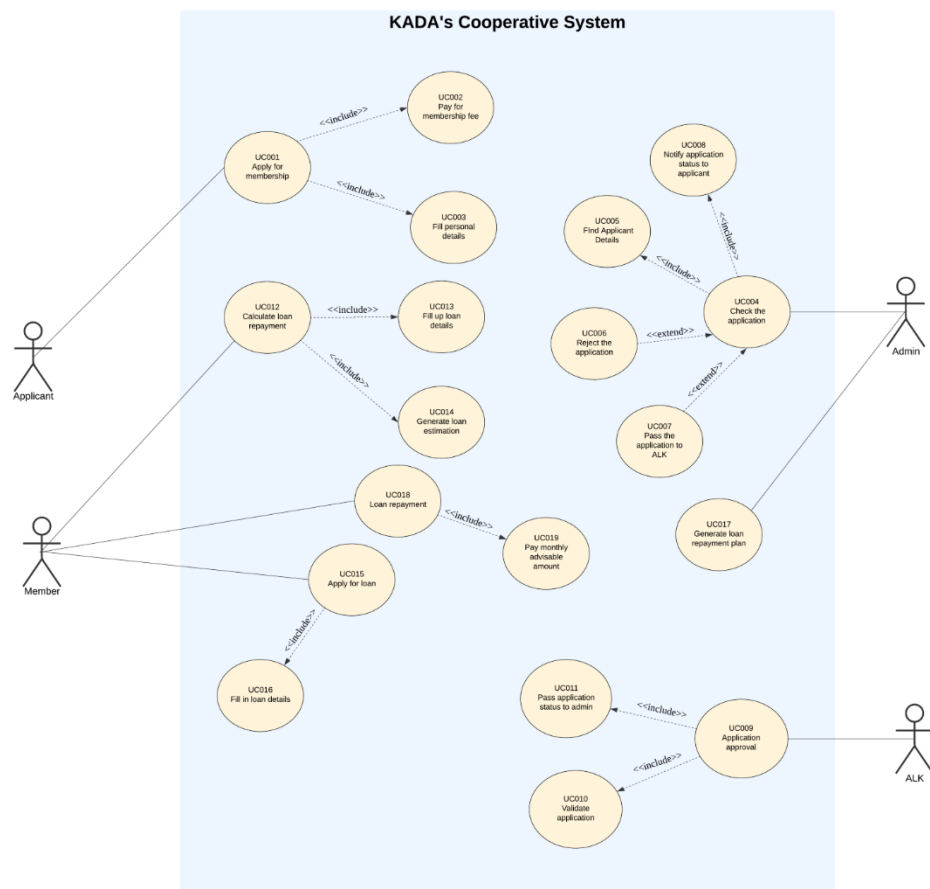
Scope:

- Online applications for membership and loans
- Centralized storage of user, membership, and loan data in a secure database
- Automated notifications to inform users about application statuses
- Role-based access for Applicants, Members, Admins, and Board Members

Boundaries:

- The system limited to handling membership and loan-related processes
- Historical manual data will not be migrated into the system

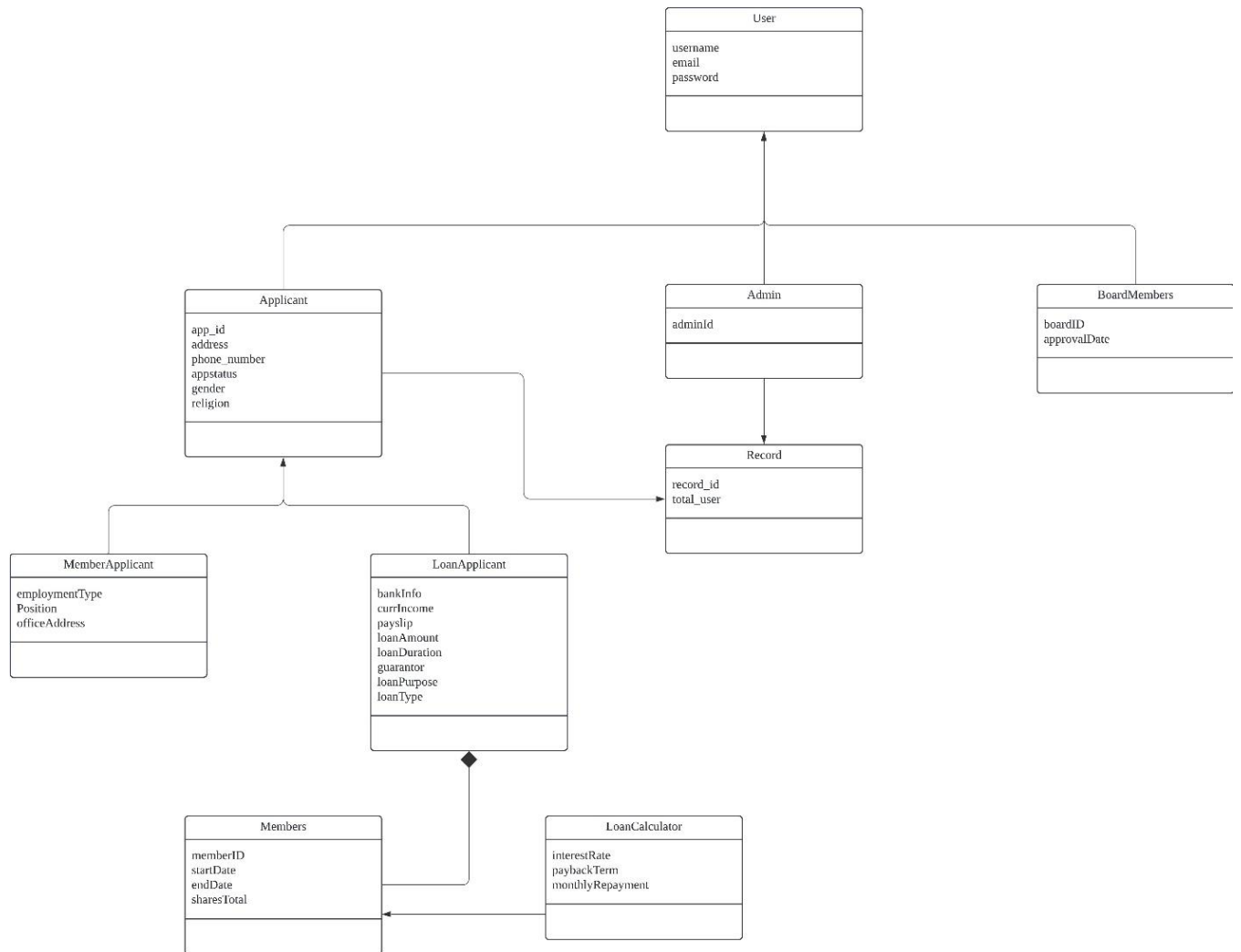
Use Case Diagram



The use case diagram includes the following interactions:

- Applicants applying for membership.
- Members applying for loans and using the loan calculator.
- Admins verifying and managing membership and loan applications and forwarding loan applications to Board Members.
- Board Members making final decisions on loan approvals or rejections.

Class Diagram



Proposed Technology:

The system will use simple and familiar tools for development:

1. Frontend:

- **HTML and Bootstrap:** To design a responsive and easy to use website layout that works on various devices

2. Backend:

- **PHP (MVC Framework):** PHP will handle the core system logic, like processing applications, communicating with the database, and managing user roles. Using the MVC (Model View Controller) framework will help organize the code and make it easier to manage.

3. Database:

- **PHPmyadmin:** The database will store all user information, membership data, and loan records securely. PHPmyadmin will be used to manage and query the database.

4. Payment Integration:

- The system will allow online payments for membership fees and loan repayments securely.

Methodology:

For this project, we have chosen to use the **Waterfall methodology**.

1. **Clear Structure:** The waterfall model provides a clear, step by step structure that aligns well with KADA's needs for digitizing their processes and improving user experience.
2. **Well Documented:** Each phase has specific deliverables, which helps track progress and ensures all requirements are met before moving to the next phase.
3. **Stakeholder Involvement:** The model allows for stakeholder review at the end of each phase, ensuring that the system meets KADA's specific requirements.

4. Glossary

List of technical terms used in the proposal, along with their definitions:

1. **Applicants:** Individuals who apply for KADA membership using the system.
2. **Members:** Approved applicants who can access additional features like loan applications.
3. **Admins:** Users responsible for verifying membership and loan applications, managing data, and ensuring system operations.
4. **Board Members (ALK):** Decision-makers who review loan applications and approve or reject them based on policies.
5. **MVC (Model-View-Controller):** A software design pattern that separates an application into three interconnected components to organize code and improve scalability.
6. **Bootstrap:** A front-end framework for creating responsive, mobile-first websites.
7. **PHP:** A server-side scripting language used to develop web applications.
8. **PHPMyAdmin:** A tool for managing MySQL databases via a graphical user interface.
9. **Centralized Database:** A single database where all system data is stored and accessed securely.
10. **Loan Calculator:** A tool within the system that helps members estimate loan repayment amounts based on variables like interest rate and duration.
11. **System Architecture (SA):** The structural design of the system, defining its components and their interactions.
12. **Requirement Specification Document:** A formal document that details the functional and non-functional requirements of the system.
13. **Waterfall Methodology:** A step-by-step approach to software development that ensures each phase is completed before moving to the next.

5. Bibliography

List of references that were used to create this proposal:

1. Sommerville, I. (2016). "Software Engineering", 10th Edition, US: Pearson.